Date: 2026-10-25

notes on papers

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(4) annotations: Bansal et al 2024-2.pdf

paper: Bansal et al. (2024)

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1

Challenges in Human-Agent Communication

#doi: arXiv:2412.10380v1

3

The broad abilities of foundation models enables the design of agents that exhibit a wide range of capabilities that often, though not always, include language understanding and generation competencies that allow people and agents to communicate through natural language.

4

The notion of grounding in communication is a central concept proposed by Clark and Brennan that embodies the collection of mutual knowledge, mutual beliefs, and mutual assumptions that are essential for communication between two people

5

the term “common ground” refers to operational alignments rather than cognitive or experiential similarities

(2) annotations: Bavaresco et al 2024.pdf

paper: @

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2

The datasets we consider cover a wide span of properties of interest, ranging from grammaticality and toxicity to coherence, factual consistency, and verbosity,

2

multiple tasks (e.g., fluency and coherence), while others are more task-specific (e.g., the success of a generated plan or the correctness of a multi-step mathematical reasoning trace

(4) annotations: Frankowsky_2022.pdf

paper: Frankowsky (2022)

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2

Frankowsky

#cite_Frankowsky_2022.pdf__

11

the date of attestation for each formation was identified.

17

Data concerning the date of attestation were not available for all 2,337 cases of ProtICCs. The year of the utterance could be identified in 772 cases (33.0%).

25

Schäfer, Roland & Bildhauer, Felix. 2012. Building Large Corpora from the Web Using a New Efficient Tool Chain. LREC: 486–493.

(3) annotations: Harding_et_al 2024.pdf

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GPT-3.5 (text-davinci-003) produces judgments about a variety of moral scenarios which correlate strongly with average human ratings

2

a successful model trained on large corpus of human text will “indirectly capture millions of human minds” (p.3). The output of such a model can then be thought of as an expression of a “modal opinion” (p.2) of the captured minds

2

recommend taking “a broadly Bayesian perspective, with data from language models providing only a small adjustment in the probability of priors

(2) annotations: interactive word embeddings for k-12 students (2022).pdf

paper: @

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5

. Expanding this panel reveals text boxes with the now classic example “king” minus “man”plus “woman”.

5

NA

marginnote4app://note/ACD02FF0-037A-402F-B1A8-C643F16750EA

(3) annotations: Kodner et al 2023.pdf

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2

Under this hypothesis, children generalize from their limited input in specific ways, navigating a constrained space of possible natural language grammars. Consequently, they do not consider all logically possible generalizations that are consistent with their linguistic experience

2

From these two claims, Piantadosi concludes that LLMs show that unconstrained learning from small data is possible

5

2.2 Small Language Models are Anything but Inevitable

(1) annotations: Luo et al 2024.pdf

paper: @

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3

have proposed a blockchain-based distributed ML called Byzantine-resilience to protect the security of the training model.

(18) annotations: Piantadosi (2024).pdf

paper: Piantadosi et al. (2024)

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6

The internal representations of words in these models are stored in a vector space, and the locations of these words include not just some aspects of meaning, but properties that determine how words can occur in sequence (e.g. syntax).

6

they are trained only on text prediction.3This means that the models form probabilistic expectations about the next word in a text and they use the true next word as an error signal to update their latent parameters

#kybernetik

6

how training a neural network on text prediction could lead it to discover key pieces of the underlying linguistic system.

#q: still doubts if just being able to reproduce lx system (grammar) yet shows an understanding of the logic (beyond probabilities) of the underlying (system). #gpt q: write a story in english based on the german / japanese / russian grammar. see: https://esteeschwarz.github.io/SPUND-LX/pages/004/lxtech003.html

7

Linguistic corpora are a low-dimensional projection of both syntax and thought, so it is not implausible that a smart learning system could recover at least some aspects of these cognitive systems from watching text alone

8

language models should be treated as bona fide linguistic theories

10

a learning mechanism configures the model itself in the space of theories in order to satisfy the desired objective function.

11

rephrasing of his (and others') comments on language learning – essentially that one should not study an unconstrained system because it will not explain why languages have the particular form that they do.9

12

In particular, we may view each model or set of modeling assumptions as a possible hypothesis about how the mind may work. Testing how well a model matches human-like behavior then provides a scientific test of that model's assumptions. This is how, for example, the field has discovered that attentional mechanisms are important for performing well

12

helping to delineate what is possible: Could it be possible to discover hierarchy without it being built in? Could word prediction provide enough of a learning signal to acquire most of grammar? Could a computational architecture achieve competence on WHquestions without movement, or use pronouns without innate binding principles?

13

Surprisingly or not, the language model vectors appear to encode at least some aspects of semantics

16

Part of making these models work well was in determining how to encode semantic properties into vectors, and in fact initializing word vectors via encodings of distribution semantics

18

Probability is central for models because a probabilistic prediction essentially provides an error signal that can be used to adjust parameters that themselves encode structure and generating processes

18

they are based in a continuous calculus that allows multiple influences to have a gradient effect on upcoming linguistic items.

18

The fact that predictions are probabilistic is useful because it means that the underlying representations are continuous and gradient

#discrete

19

Continuity is important because it permits the models to use gradient methods – essentially a trick of calculus – to compute what direction to change all the parameters in order to decrease error the fastest.

19

prevalence of “relaxation” methods in numerical computing, where an optimization problem with hard constraints is often best solved via a nearby soft, continuous optimization problem

19

The success of gradient models over deterministic rules suggests that quite a lot of language is based on gradient computation

22

aux-inversion

(1) annotations: Wilkinson et al. 2016.pdf

paper: @

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Findability, Accessibility, Interoperability, and Reusability

(21) annotations: Zinsmeister 2013.pdf

paper: Zinsmeister (2013)

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1

Heike Zinsmeister

#cite_Zinsmeister 2013.pdf__

2

Our corpus-based study is grounded in the approach of Distributional Semantics, which assumes that the meaning of a word can be modeled by collecting information about its distribution in a corpus

2

we distinguish between three different semantic compound types. The first type is traditionally referred to as a determinative (or endocentric) compound (cf. Olsen 2000). Its meaning is compositionally derived from the meanings of its parts: The meaning of the head noun denotes the type of referent, and the meaning of the modifier noun modifies this in some semantic relation to the head noun, such that it further specifies or modifies the meaning of the head noun.

2

The distributional information is collected by examination of co-occurring words, structures, or relations. The sum of this contextual information can be viewed as a kind of mold, a container that shapes the meaning of the word in a characteristic fashion.

3

The final semantic type is traditionally called the copulative (or dvandva) compound

3

includes all noun-noun compounds that are lexicalized such that their meanings are opaque with respect to the meanings of their parts.(..)(..)They are semantically

3

is the least frequent type of compound in German. In Fanselow's (1981b) definition, the meaning of a copulative compound is created by applying the basic relation 'AND' to the meanings of its parts

3

The second semantic type is traditionally referred to as the possessive (also bahuvrihi or exocentric) compound.

4

the distributions of the target nouns are vectors.

5

In such a vector space, each context word corresponds to a dimension, and the distribution of a target word is a vector in this multidimensional space. The similarity between two words can then be measured in terms of the distance between their vectors in the space

5

The similarity of two distributions is interpreted as the distance between their vectors. A common way to measure this distance is to calculate the cosine of the angle between the two vectors (e.g., Erk 2012: 637). If two vectors point in the same direction, meaning that the angle between them is 0°, then the cosine will be 1. Otherwise, the cosine will be less than 1; it will be - 1 if the two vectors point in exactly opposite directions

5

The basic elements in the standard model are the co-occurrence frequencies of context words. Context is defined as a window on the token string.

6

The Kullback-Leibler divergence

6

skew divergence as a variant of the KullbackLeibler divergence. Instead of calculating the divergence using the observed probability distributions, this function smoothes the values of q, meaning that q will be positive for all dimensions for which p has a positive value

6

Lee (1999, 2001) has introduced skew divergence

7

extracted all words that were annotated as common nouns in the corpus and analyzed them further using SMOR (Schmid et al. 2004

7

skewed distribution of head family size

what?

8

skew divergence distributions D. These scores range from a minimum of 0.093, observed for the divergence of a head from its compound (cf. leftmost boxplot D(c(..)(..)h)), to a maximum of 2.293, observed for the divergence of a compound from its modifier

8

The similarity of head and compound will exceed that of modifier and compound:

8

The similarity of head and compound will not exceed the similarity of modifier and compound:

8

the compound will have a high similarity to both head and modifier:

Bansal, Gagan, Jennifer Wortman Vaughan, Saleema Amershi, Eric Horvitz, Adam Fourney, Hussein Mozannar, Victor Dibia, and Daniel S. Weld. 2024. *Challenges in Human-Agent Communication*. arXiv. https://doi.org/10.48550/arXiv.2412.10380.

Frankowsky, Maximilian. 2022. *Extravagant Expressions Denoting Quite Normal Entities: Identical Constituent Compounds in German*. John Benjamins Publishing Company. https://www.degruyterbrill.com/document/doi/10.1075/slcs.223.07fra/html.

Piantadosi, Steven T., Stefan Müller, Moshe Poliak, Geoffrey K. Pullum, Robert D. Van Valin Jr, Dafydd Gibbon, Iris Berent, et al. 2024. "Modern Language Models Refute Chomsky's Approach to Language." In *From Fieldwork to Linguistic Theory*, 353–415. Language Science Press. https://langsci-press.org/catalog/book/434.

Zinsmeister, Heike. 2013. "Corpus–Based Modeling of the Semantic Transparency of Noun–Noun Compounds." In *A Festschrift for Susan Olsen*, edited by Holden Härtl, 303–21. Berlin: Akademie Verlag. https://doi.org/doi:10.1524/9783050063799.303.

